

SCIENCE MASTER X

Complete Academic Study Notes • Class 10 Science

CHAPTER 09

Light – Reflection and Refraction

(Full Chapter)

Laws of Reflection • Spherical Mirrors & Ray Diagrams • Mirror Formula & Magnification • Refraction & Snell's Law • Refractive Index • Lenses & Image Formation • Lens Formula & Power of Lens

Target Level: CBSE Class 10

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Chapter Overview & Basic Concepts

Light is a form of electromagnetic radiation that enables vision. It travels in straight lines (rectilinear propagation of light) and exhibits phenomena such as reflection, refraction, and dispersion.

SECTION 1: REFLECTION OF LIGHT & SPHERICAL MIRRORS

1.1 Laws of Reflection

- **First Law:** The angle of incidence ($\angle i$) is always equal to the angle of reflection ($\angle r$). That is, $\angle i = \angle r$.
- **Second Law:** The incident ray, the normal to the reflecting surface at the point of incidence, and the reflected ray all lie in the same plane.

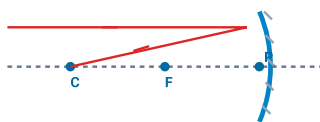
1.2 Types of Spherical Mirrors

- **Concave Mirror:** A spherical mirror whose reflecting surface is curved inwards towards the center of the sphere. (Converging Mirror)
- **Convex Mirror:** A spherical mirror whose reflecting surface is curved outwards. (Diverging Mirror)

1.3 Important Terminology for Mirrors

- **Pole (P):** The center point of the reflecting surface of a spherical mirror.
- **Center of Curvature (C):** The center of the sphere of which the mirror forms a part.
- **Radius of Curvature (R):** The radius of the sphere of which the mirror forms a part.
- **Principal Focus (F):** The point on the principal axis where rays parallel to the principal axis meet (concave) or appear to diverge from (convex).
- **Relationship:** For spherical mirrors of small aperture, $R = 2f$.

 **Diagram: Reflection Through Concave Mirror**



NCERT High-Yield Insight: A concave mirror forms a virtual, erect, and magnified image ONLY when the object is placed between the Pole (P) and Focus (F). This unique property makes it suitable for shaving and dental mirrors!

SECTION 2: MIRROR FORMULA, SIGN CONVENTION & APPLICATIONS

2.1 New Cartesian Sign Convention

- The Pole (P) of the mirror is taken as the origin.
- Distances measured in the direction of incident light are positive (+).
- Distances measured opposite to the direction of incident light are negative (-).
- Object distance (u) is ALWAYS negative.
- Focal length (f) of Concave Mirror is ALWAYS negative (-). Focal length (f) of Convex Mirror is ALWAYS positive (+).

Important Mirror Formulae:

• Mirror Formula: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

• Magnification (m): $m = \frac{h}{h} = -\frac{v}{u}$

(Note: If m is negative, image is real and inverted. If m is positive, image is virtual and erect.)

2.2 Summary of Mirror Applications

Mirror Type	Applications / Uses	Reason
Concave Mirror	Torches, searchlights, vehicle headlights, dentist mirrors, solar furnaces.	Produces powerful parallel light beams when light source is at focus; forms enlarged virtual image when object is close.
Convex Mirror	Rear-view mirrors in vehicles, street light reflectors, shop security mirrors.	Always forms an erect, diminished image and provides a much wider field of view.

Diagram: Image Formation by Convex Mirror (Rear-view)



SECTION 3: REFRACTION OF LIGHT & REFRACTIVE INDEX

3.1 Refraction of Light

The bending of light rays when they pass obliquely from one transparent medium to another medium due to a change in the speed of light.

Rare to Denser Medium → Light bends TOWARDS the normal (Speed decreases)
Denser to Rarer Medium → Light bends AWAY from the normal (Speed increases)

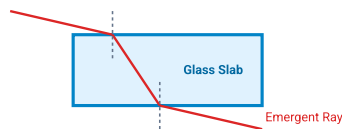
3.2 Laws of Refraction & Snell's Law

- **First Law:** The incident ray, the refracted ray, and the normal to the interface of two transparent media at the point of incidence all lie in the same plane.
- **Second Law (Snell's Law):** The ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant for a given pair of media and given color of light. $\frac{\sin i}{\sin r} = \text{constant} \ (n_{21})$

3.3 Refractive Index (n)

- **Refractive Index (n_{21}):** Ratio of speed of light in medium 1 (v_1) to speed of light in medium 2 (v_2). $n_{21} = \frac{v_1}{v_2}$.
- **Absolute Refractive Index (n):** Ratio of speed of light in vacuum/air ($c = 3 \times 10^8 \text{ m/s}$) to speed of light in the given medium (v). $n = \frac{c}{v}$.

Diagram: Refraction Through Rectangular Glass Slab



SECTION 4: SPHERICAL LENSES, LENS FORMULA & POWER

4.1 Types of Lenses

- **Convex Lens:** Thicker at the middle than at the edges. Converges light rays.
- **Concave Lens:** Thicker at the edges than at the middle. Diverges light rays.

Important Lens Formulae & Power:

- Lens Formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$
- Magnification (m): $m = \frac{h}{h} = \frac{v}{u}$
- Power of Lens (P): $P = \frac{1}{f \text{ (in meters)}}$ (SI Unit: **Diopetre, D**)
(1 Diopetre is the power of a lens whose focal length is 1 meter.)

4.2 Comparison Table: Mirror vs. Lens Sign Convention

Quantity	Concave Mirror	Convex Mirror	Convex Lens	Concave Lens
Focal Length (f)	Negative ($-$)	Positive ($+$)	Positive ($+$)	Negative ($-$)
Power (P)	—	—	Positive ($+$)	Negative ($-$)
Object Distance (u)	Negative ($-$)	Negative ($-$)	Negative ($-$)	Negative ($-$)

? Important Practice Questions

Q1: A convex mirror used for rear-view on an automobile has a radius of curvature of 3.00 m. If a bus is located at 5.00 m from this mirror, find the position and nature of the image.

Ans: $R = +3.00 \text{ m} \implies f = +1.50 \text{ m}$; $u = -5.00 \text{ m}$

Using mirror formula: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{1.50} - \frac{1}{-5.00} = \frac{1}{1.50} + \frac{1}{5.00} = \frac{13}{15} \implies v = +1.15 \text{ m}$

Image is virtual, erect, and formed behind the mirror.

Q2: State Snell's law of refraction of light.

Ans: Snell's law states that the ratio of the sine of the angle of incidence ($\sin i$) to the sine of the angle of refraction ($\sin r$) is constant for a given pair of media: $\frac{\sin i}{\sin r} = \text{constant}$ (n_2).

Q3: Find the focal length of a lens of power -2.0 D. What type of lens is this?

Ans: $f = \frac{1}{P} = \frac{1}{-2.0 \text{ D}} = -0.5 \text{ m} = -50 \text{ cm}$. Negative power and focal length indicate that it is a **Concave Lens**.

Q4: Define 1 Diopetre of power of a lens.

Ans: 1 Diopetre is defined as the power of a lens whose focal length is exactly 1 meter ($1 \text{ D} = 1 \text{ m}^{-1}$).

